Rotaract Club of Xiamen University Malaysia | Rotary Club of Kuala Lumpur Dynamics | Rotary Club of Tun Razak Exchange

SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

Instagram: @sdc_xmumhack ***Email:*** sdc.xmumhack@gmail.com

SDC HACKATHON 2026

Phase 1: Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club
Submission Deadline: 25 October 2026, 07:59:59 MYT

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

1. Team & Project Overview

Team Name	UTeMku Cemerlang
Selected Domain	Domain 2 - Orbyte Smart SME Challenge
Selected SDG Track (1–17)	SDG 8 — Decent Work and Economic Growth
Project / Solution Name	PaySah
Name (Leader)	Chong Kai Li
Email & Phone (Leader)	kailichong148@gmail.com 0146312883
University Affiliation(s)	Universiti Teknikal Malaysia Melaka
Number of Team Members	4

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Micro and small merchants receiving payment screenshots must match each receipt to an order and check transaction records. Downloading images, uploading them into a checking tool, searching orders and recording decisions creates repeated administration and can delay fulfilment.

A screenshot alone does not prove settlement. Similar amounts, missing references, partial payments, scheduled transfers and reused screenshots create uncertainty and increase the risk of incorrect matching.

PaySah targets this receipt-to-order verification workload. Customers submit screenshots through Telegram or Google Form; merchants should not need to download and re-upload them. The intended benefit is less manual work with clear evidence and human control.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

PaySah turns payment-screenshot checking into one workflow. Cashiers or Owners generate a one-time link for an order; the customer sends the screenshot through Telegram or Google Form, so the merchant no longer downloads and re-uploads images. The Telegram bot and Google Form are connected to the intake backend, and the complete receipt-intake workflow is deployed on Replit.

A vision-language model (Qwen2.5-VL-7B on a RunPod GPU) reads amount, account, time, reference and transfer status. Tesseract.js CPU OCR reads the same image as a cross-check and is the fallback when the GPU is unavailable. Deterministic rules then compare the receipt with the order and with team-created synthetic bank transactions. Live banking is outside scope.

The review screen shows extracted fields, evidence and exception reasons. Unreadable fields, GPU/CPU disagreements, pending or scheduled transfers, wrong accounts, duplicate screenshots and missing bank records stay unpaid for human review. Cashiers can confirm only clean matches up to RM500; larger amounts, partial payments and exceptions go to the Owner. A partial payment cannot be marked paid or released. Corrections, approvals and status changes are logged.

The prototype is deployed on Replit (https://paysah.replit.app) with React, Node.js and PostgreSQL, applying the GenAI, database and cloud deployment workshop topics. A simulated Orbyte interface lets the same check serve sales orders and booking deposits; no live Orbyte connection is claimed.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Implemented stack: TypeScript; React 19 and Vite; Node.js 24 and Express 5; Zod validation; PostgreSQL 17 (pg with SQL migrations); Multer and Sharp for image upload and preprocessing; Tesseract.js 7 CPU OCR with deterministic field parsers; Qwen2.5-VL-7B-Instruct served by vLLM on RunPod Serverless; node:test and Playwright for testing.

System flow: Telegram / Google Form order link → intake API → private image storage with SHA-256 hash → field extraction (RunPod GPU model, CPU OCR cross-check and fallback) → rule checks against the order and synthetic bank transactions → Cashier / Owner review → payment status, separate release action and activity log.

Controls: server-side Cashier / Owner roles; amounts above RM500, partial payments and exceptions require the Owner; exact-duplicate image detection; one bank transaction cannot be allocated twice; idempotent, version-checked updates. Orbyte interfaces are simulated.

Sponsor platforms: Yes. Replit hosts the deployed web app, API, OCR worker, PostgreSQL database and private Object Storage (https://paysah.replit.app). RunPod Serverless (24 GB GPU) runs the vision model. On 36 team-made synthetic receipts it read 248/252 fields versus 190/252 for CPU OCR alone. Because the model sometimes guessed unreadable values, every GPU/CPU disagreement is routed to the Owner instead of being auto-accepted.

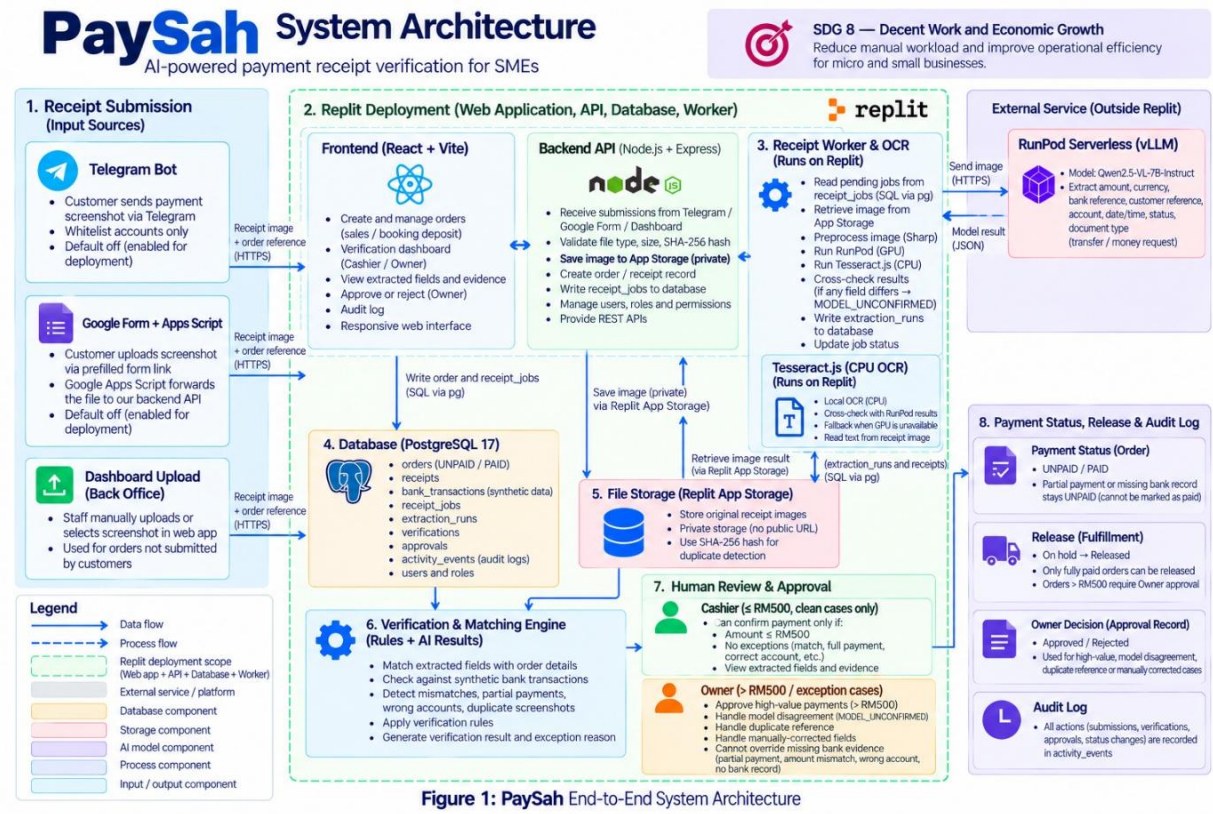


Figure 1: PaySah End-to-End System Architecture

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

Empathise and define: we mapped the merchant's steps (receive screenshot, download, upload, find order, check bank record, decide) and defined the task as collection, matching and verification. Mentor feedback showed collection was a major manual step, so we built automatic intake through Telegram and Google Form.

Prototype and test: we built a review screen with extracted fields, evidence and exception reasons, and tested 36 synthetic receipts. CPU OCR alone read 190/252 fields; a RunPod GPU model read 248/252, but sometimes guessed hidden values, so any GPU/CPU disagreement now goes to the Owner.

Deploy: the prototype already runs on Replit. Before Phase 2 we will compare manual processing with PaySah on identical synthetic cases: normal, partial, wrong-account, scheduled, duplicate, missing-transaction and equal-value orders, plus intake and OCR failures. We will record handling time, manual steps and matching errors, and use new-user feedback to refine the interface. Merchant validation remains pending.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	Primary: SDG 8.2, productivity through technological upgrading and innovation. Supporting alignment: SDG 8.3, entrepreneurship and micro and small enterprise growth. These are operational contributions, not national indicator measurements. Source: United Nations, Goal 8, https://sdgs.un.org/goals/goal8 (accessed 24 October 2026).
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	Workflow baseline: 2 merchant file-handling actions per receipt: download and upload. This counts the workflow being replaced; it is not an industry average. Manual checking time and error rates remain unmeasured and will be established on synthetic cases.
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	Primary: merchant download/upload actions per collected receipt. Supporting: median staff handling time, incorrect order matches and unresolved cases routed to review. Time reduction = (manual median - PaySah median) / manual median x 100%.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	Proposed: 100% reduction in merchant file-handling actions, from 2 to 0, for successfully collected receipts from the connected channel(s); at least 30% lower median staff handling time on identical synthetic cases. Targets are unvalidated. Track matching errors alongside time savings. No environmental savings are quantified.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

- [x] All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- [x] The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- [x] The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- [x] All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☑ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☑ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
React, React DOM, Vite, Express, Multer, Zod, pg, concurrently, tsx	Code libraries	MIT
TypeScript, Sharp, Tesseract.js, Playwright	Code libraries	Apache-2.0
Node.js	Runtime	MIT
PostgreSQL; embedded-postgres (local development only)	Database	PostgreSQL License; MIT
@replit/object-storage	Code library (Replit SDK)	MIT
@tesseract.js-data/eng (English OCR language data)	Model data	As declared by package (MIT)
Qwen/Qwen2.5-VL-7B-Instruct	Pre-trained vision-language model	Apache-2.0
vLLM (RunPod Serverless worker)	Inference server	Apache-2.0
Team-created synthetic orders, receipts and bank transactions	Synthetic dataset	Team-created; no real customer records

Exact versions are pinned in package-lock.json. The project is released under Apache License 2.0.

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☑ No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.



[Signature]

[24/106/202]